William Lo Channiko

APPLIED AI ENGINEER

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Applied AI engineer with 1 year of full-time software engineering experience. Builds AI assistants, custom MCP servers, and reusable business skills for sales, service operations, and administration. Combines Python/API development, OpenAI and Ollama integration, access controls, testing, monitoring, and AWS/Azure deployment validation.

TECHNICAL SKILLS

AI & MCP: Custom MCP servers, reusable business skills, OpenAI API, LangGraph, RAG, embeddings, structured outputs, Mistral OCR

Local AI: Ollama setup and local inference integration; Docker Compose environments; provider evaluation

Software & integrations: Python, TypeScript, SQL, FastAPI, React, Next.js; REST APIs, PostgreSQL/pgvector, Odoo, ERPNext, automation scripts

Security & reliability: RBAC, tenant isolation, human approval, audit trails, idempotency, recovery; Pytest, Vitest, Playwright, CloudWatch monitoring

Cloud & infrastructure: AWS (ECS/Fargate, RDS, SQS, S3, Lambda); Azure (Container Apps, Blob Storage, Service Bus, PostgreSQL); Docker, CI/CD

MCP & BUSINESS AUTOMATION

ConnectWise Service Operations MCP | GitHub Repository *Oct 2026*

- Built a service-operations MCP server aligned with a publicly documented ConnectWise PSA API subset, exposing six tools for scoped ticket context, internal notes, and evidence-backed time entries with separate-user approval.
- Validated 62 automated tests, 11 MCP protocol scenarios, and eight browser checks; completed 135 synthetic workload tasks, including 33 verified writes, with zero observed errors or duplicate effects against a stateful PSA simulator.
- Engineered the platform with TypeScript, FastAPI, PostgreSQL, Docker, and optional OpenAI/Ollama assistance; implemented a responsive operator workspace, tenant isolation, audit trails, durable synchronization, and read-back verification with unknown-outcome recovery.

Odoo Business Operations MCP Server | GitHub Repository *Oct 2026*

- Built an AI-assisted Odoo operations platform with four reusable business skills and ten scoped MCP tools for customer research, quotation preparation, CRM follow-ups, and write reconciliation, enforcing tenant isolation and independent human approval.
- Validated 124 synthetic workload tasks, including 31 quotation writes with exactly one Odoo order per operation across replay checks; passed 13 browser acceptance checks covering approval separation, cross-company access denial, and recovery after a lost execution response.
- Engineered the platform with TypeScript, Python/FastAPI, PostgreSQL, Odoo, and Docker Compose; delivered a responsive web workspace with source-bound approvals, idempotent execution, verified business receipts, and persistent worker recovery.

AI Service Desk & Ticket Operations Assistant | GitHub Repository *Oct 2026*

- Built an AI-assisted service desk platform that converts Jira tickets into evidence-backed recommendations and versioned action proposals, with human approval for access grants, service recovery, and related-ticket operations.
- Validated AI triage on 16/16 held-out synthetic cases and exercised 48 synthetic HTTP workflows; verified platform behavior through 183 Python tests, 8 MCP tests, and 21 JavaScript checks, with automated regression and secret scanning in GitHub Actions.
- Engineered the platform with Python, PostgreSQL, TypeScript/MCP, OpenAI, Jira Service Management, and Keycloak; implemented tenant isolation, payload-bound approvals, durable operation identities, and target verification to reconcile uncertain outcomes and prevent duplicate effects.

AI PROJECTS

Case Resolution Copilot | GitHub Repository *Jul-Sep 2026*

- Built a policy-governed AI case-resolution system that converts fragmented evidence and versioned policies into review-ready Decision Briefs, with deterministic risk controls and human approval for consequential actions.
- Reduced raw median workflow time by 83.7% (582s to 95s) in a developer-operated benchmark across three matched synthetic cases; validated critical behavior through 20/20 regression observations, a 3/3 OpenAI canary, and 4/4 PostgreSQL workflow scenarios.
- Engineered the platform with FastAPI, PostgreSQL/pgvector, LangGraph, Celery, and Next.js; strengthened workflow reliability through idempotent actions, audit trails, and worker recovery. Live-validated a reproducible AWS deployment with CloudWatch monitoring across ECS/Fargate, RDS, SQS, S3, Lambda, CloudFront, IAM, and Secrets Manager.

Invoice Review | GitHub Repository *Jul-Aug 2026*

- Built an AI invoice-to-ERP workflow that turns unstructured documents into validated, review-ready invoice data, combining OCR extraction, deterministic validation, and human approval before ERPNext export.
- Reduced median invoice-to-ERP draft time by 68% (153s to 49s) across six paired synthetic invoices; achieved the expected outcome in 10/10 test cases, compared with 9/10 through manual ERPNext entry.
- Integrated React, FastAPI, Mistral OCR, OpenAI, and ERPNext; validated an end-to-end Azure deployment using Container Apps, private Blob Storage, Service Bus, and PostgreSQL, processing a synthetic invoice from upload through the review queue.

PROFESSIONAL EXPERIENCE

PT Dover Chemical

Full-Stack Software Engineer Intern | *Jakarta, Indonesia* *Jan 2025-Jan 2026*

- Owned end-to-end development of an offline-first Draft Sales Order application, from requirements gathering with IT supervisors and sales users through implementation, integration, and testing; built local persistence, background synchronization, caching, retries, and attachment handling across 38 API operations and 15 Room entities/DAOs.
- Independently designed and built a Customer Management System end to end around a 6,656-record customer master dataset, translating sales and administrative requirements into validated imports, reporting, audit trails, and role-based approval, rejection, and reopen workflows across 15 data domains.

EDUCATION

Multimedia Nusantara University, Tangerang, Indonesia 2022-2026

Bachelor's Degree in Informatics (S.Kom.) | Graduated with Distinction | GPA: 3.82/4.00

Thesis: Fine-tuned two IndoBERT classifiers on 1,822 Indonesian social-media opinions; achieved test macro-F1 of 0.981 for sentiment and 0.870 for aspect classification, with inter-annotator Cohen's kappa of 0.954 and 0.900 on a 200-opinion reliability sample.

Honors: Academic Achievement Scholarship, 2024